A Two Dimensional Model for CO2 Uptake and Tank pH Evolution in a Recirculating Hollow Fiber Membrane Contactor

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Abstract

This report presents a model that connects contactor geometry, gas and liquid flow, initial absorbent composition, and transport properties to carbon dioxide uptake and tank pH evolution. Radial and axial transport in a hollow fiber is coupled to carbonate equilibrium, sequential shell-side gas balances, and a well-mixed recirculation tank. Dissolved inorganic carbon and alkalinity serve as transported variables, with species concentrations and pH reconstructed from the local chemical state. A 20 × 20 fiber array is represented by 20 gas-flow rows with identical fibers within each row. Under the prescribed reference conditions, the tank pH decreases from 13.00 to 10.236 over $48\,\mathrm{h}$ while the liquid absorbs an additional $34.505\,\mathrm{mmol}$ of CO_2. The $24\,\mathrm{h}$ spatial state shows small pH variations and a 0.31% difference in mean absorption flux between the first and last rows. With the same alkalinity and an initial DIC of $50\,\mathrm{mol\,m^{-3}}$, additional uptake is $13.587\,\mathrm{mmol}$. Both cases satisfy the specified conservation and refinement checks. The predictions use ideal chemistry and an uncalibrated external transfer coefficient.

1. Introduction and Objectives

1.1. Prediction Objective

The objective is to predict how CO_2 absorption changes the liquid inventory and pH in a recirculating hollow-fiber membrane contactor. Prescribed geometry, operating conditions, initial dissolved inorganic carbon (DIC), alkalinity, and transport properties determine the predicted tank pH, cumulative carbon uptake, absorption rate, and internal concentration fields. These outputs connect a device-scale response to transport and chemical redistribution within the fibers.

Three questions guide the report: how the reference device evolves in time, which spatial distributions explain its response, and how the response changes when the liquid is initially partly carbonated.

1.2. Scope of the Model

The liquid chemistry is restricted to the potassium hydroxide–carbonate–water system. The model assumes ideal local equilibrium, prescribed temperature and diffusivities, electroneutral bulk liquid, and zero radial electric current. A quasi-steady fiber calculation supplies the instantaneous uptake to a time-

dependent mixed-tank balance. Individual fiber maldistribution, finite hydration kinetics, activity corrections, wetting, and transient storage within the fibers are not resolved.

Reaction-invariant formulations and local-equilibrium reductions have a well-established role in aquatic chemical modeling [1]. Reactive-transport implementations also show why transport must be transformed consistently when chemical species are reconstructed from equilibrium variables [2]. The work presented here applies these ideas to the fiber geometry, makes the charge constraint explicit, and connects the resulting equations to a conservative discretization.

Appendix A derives the reduced model, and Appendix B documents numerical verification.

2. Model Development

2.1. Contactor Configuration and Assumptions

The array contains 400 fibers in 20 gas-flow rows of 20 fibers each. Liquid from the mixed tank enters all fibers in parallel at equal flow per fiber, while gas passes successively through the rows. Fibers within a row share identical conditions, so one representative fiber is solved per row. Figure 1 shows the device arrangement and the resolved lumen coordinates.

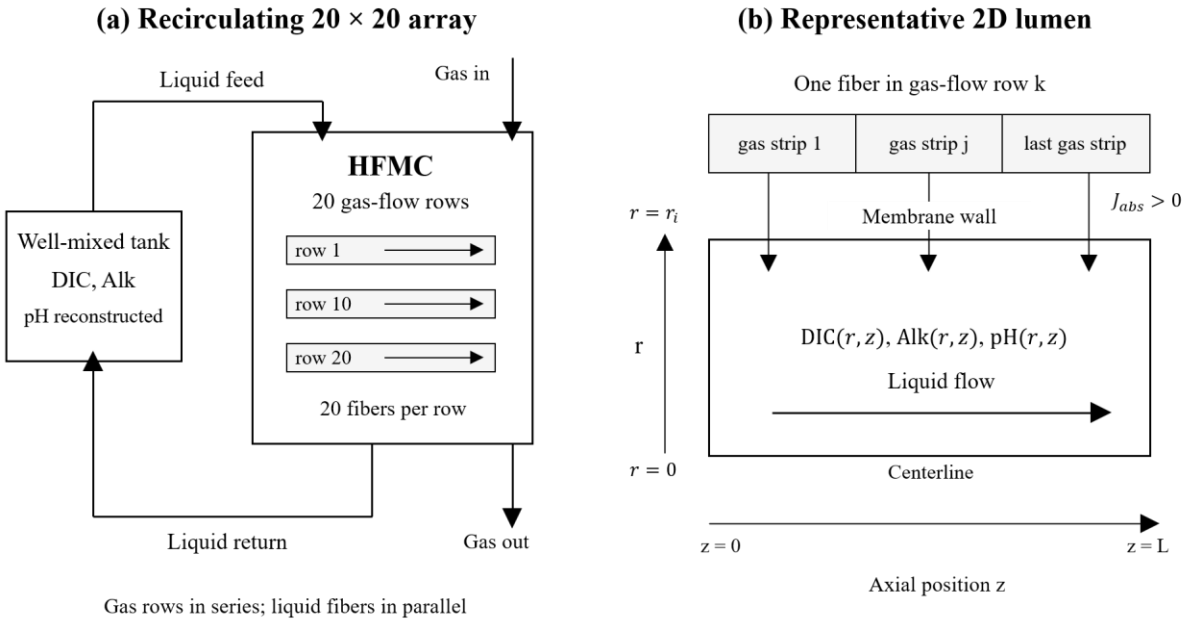


Figure 1. Recirculating 20 × 20 Hollow-Fiber Array and Two-Dimensional Model Coordinates. (a) Liquid from the mixed tank enters 400 fibers in parallel, while gas passes sequentially through 20 rows of 20 fibers. (b) A representative lumen is resolved in axial z and radial r; the arrows denote CO_2 transfer into the liquid. Gas composition is updated by row and axial strip.

Consider a cylindrical lumen of inner radius r_i and active length L. The membrane outer radius is r_o. The axial coordinate z follows liquid flow and the radial coordinate r increases from the centerline toward the membrane. The axisymmetric domain and fully developed velocity are

$$0 \le r \le r_i, \quad 0 \le z \le L, \quad v_z(r) = 2\bar{v}\left[1 - \left(\frac{r}{r_i}\right)^2\right].$$

The volumetric flow through one fiber is $q_f = \pi r_i^2 \bar{v}$. Temperature, liquid density, viscosity, and species diffusivities are prescribed. Radial fluid velocity is zero. The slender-fiber approximation retains axial advection and radial nonadvective transport while neglecting axial diffusion and axial electromigration.

Table 1. Model Assumptions and Their Consequences.

Assumption	Consequence	Applicability condition
Axisymmetric developed laminar flow	Prescribed parabolic axial velocity	Entrance effects and azimuthal variations are negligible
Constant temperature and dilute ideal chemistry	Concentration-form equilibrium constants	Activity and composition-dependent property effects are omitted
Fast carbonate and water reactions	Algebraic local-equilibrium closure	Chemical relaxation is fast relative to resolved transport
Electroneutral bulk liquid	Potassium equals carbonate-system alkalinity	Only the stated ions are present; electric double layers are unresolved
Radial ionic transport with no imposed current	Radial electric field follows from zero charge flux	Axial nonadvective current is negligible at the retained order
Selective neutral carbon transfer	DIC enters through the membrane; alkalinity has zero net membrane flux	No ion permeation or other alkalinity source
Quasi-steady fiber with a mixed tank	Fiber storage is omitted in the coupling model	Fiber relaxation is short relative to tank-state variation

The neutral dissolved component, denoted $\mathrm{CO_2^*}$, lumps dissolved carbon dioxide and carbonic acid consistently with the chosen first dissociation constant. The membrane condition is an effective condition for the reduced equilibrium model; its relationship to species-resolved conditions is discussed in Section 2.4.

2.2. Chemical State and pH Calculation

Let the six-species concentration vector include potassium explicitly:

$$\mathbf{c} = (c_0, c_1, c_2, c_{\mathrm{OH}}, h, c_K)^T.$$

The entries refer, respectively, to $\mathrm{CO_2^*}$, $\mathrm{HCO_3^-}$, $\mathrm{CO_3^{2-}}$, $\mathrm{OH^-}$, $\mathrm{H^+}$, and $\mathrm{K^+}$. Concentrations in dimensional equations are expressed in $\mathrm{mol\,m^{-3}}$. Define dissolved inorganic carbon DIC and alkalinity Alk by

$$\mathrm{DIC} = c_0 + c_1 + c_2, \quad \mathrm{Alk} = c_1 + 2c_2 + c_{\mathrm{OH}} - h.$$

In matrix form,

$$\mathbf{y} = \begin{pmatrix} \mathrm{DIC} \\ \mathrm{Alk} \end{pmatrix} = B\mathbf{c}, \quad B = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & -1 & 0 \end{pmatrix}.$$

Internal reactions conserve DIC and alkalinity, so their sources cancel in the two total balances. The separate local-equilibrium assumption reconstructs species from these totals; Appendix A.1 gives the stoichiometric derivation.

Use positive concentration-form constants K_1, K_2, and K_w at the prescribed temperature. Product-form equilibrium relations, which remain meaningful when total carbon is zero, are

$$hc_1 = K_1 c_0, \quad hc_2 = K_2 c_1, \quad hc_{\mathrm{OH}} = K_w.$$

Define the denominator and species fractions as

$$d(h) = h^2 + K_1 h + K_1 K_2, \quad (\alpha_0, \alpha_1, \alpha_2) = \frac{(h^2, K_1 h, K_1 K_2)}{d(h)}.$$

The carbonate species are $c_k = \mathrm{DIC}\alpha_k$ for $k = 0,1,2$, and hydroxide is $c_{\mathrm{OH}} = K_w/h$. The hydrogen concentration follows from

$$g(h; \mathrm{DIC}, \mathrm{Alk}) = \mathrm{DIC}\frac{K_1 h + 2K_1 K_2}{h^2 + K_1 h + K_1 K_2} + \frac{K_w}{h} - h - \mathrm{Alk} = 0.$$

The charge-number vector and electroneutrality condition give

$$\boldsymbol{\zeta} = (0, -1, -2, -1, 1, 1)^T, \quad \boldsymbol{\zeta}^T \mathbf{c} = c_K - \mathrm{Alk} = 0.$$

Thus $c_K = \mathrm{Alk}$ and the physical domain for the potassium-only electrolyte is $\mathrm{DIC} \geq 0$ and $\mathrm{Alk} > 0$. Within this domain, the equilibrium state is a function $\mathbf{c} = \mathbf{c}(\mathbf{y})$, with Jacobian $G = \partial\mathbf{c}/\partial\mathbf{y}$.

The ideal concentration-based pH is $\mathrm{pH} = -\log_{10}(h/c^\circ)$, where $c^\circ = 1000\ \mathrm{mol\,m^{-3}}$. The equilibrium constants use concentration units at the prescribed temperature.

The prescribed initial DIC and alkalinity determine the initial pH. Appendix A.2 proves uniqueness of the positive hydrogen-ion root and gives its derivatives.

2.3. Two Dimensional Liquid Transport

Section 2.2 determines the species concentrations and pH from the local DIC and alkalinity. Their spatial distributions require a transport law. Since $\mathbf{c} = \mathbf{c}(\mathbf{y})$, the chemical Jacobian G converts gradients of the two totals into species concentration gradients:

$$\partial_r \mathbf{c} = G \partial_r \mathbf{y}.$$

Unequal ionic diffusivities can produce a nonzero charge flux under independent diffusion. The Nernst–Planck law [3] includes the electric field needed for zero radial current under the slender-fiber approximation of Section 2.1. Define

$$D = \mathrm{diag}(D_i), \qquad b_i = \zeta_i D_i c_i, \qquad \kappa = \sum_i {\zeta_i}^2 D_i c_i > 0.$$

Here D_i is the diffusivity of species i. Eliminating the electric field gives the species transport operator $\mathcal{L}$ and the vector of six radial species fluxes $\mathbf{j}_r$:

$$\mathcal{L} = D - \frac{\mathbf{b}\boldsymbol{\zeta}^T D}{\kappa}, \qquad \mathbf{j}_r = -\mathcal{L} G \partial_r \mathbf{y}.$$

The DIC and alkalinity fluxes are obtained by applying the same matrix B that defines these totals from the species concentrations:

$$\mathbf{J}_{\mathbf{y},r} = B\mathbf{j}_r = -B\mathcal{L}G\partial_r \mathbf{y} = -M(\mathbf{y})\partial_r \mathbf{y}.$$

The resulting 2×2 matrix $M(\mathbf{y}) = B\mathcal{L}G$ combines local equilibrium, species transport, and projection onto the two totals. Its composition dependence and cross terms are retained. Combining axial advection with radial flux conservation gives

$$\partial_t \mathbf{y} + v_z(r)\partial_z \mathbf{y} = \frac{1}{r}\partial_r [r M(\mathbf{y}) \partial_r \mathbf{y}].$$

For each tank state, the device solver omits the fiber time derivative. The mixed-tank balance in Section 2.5 supplies the time-dependent composition and pH, while the inlet and membrane conditions in Section 2.4 close the fiber problem. Appendix A.3 derives the electrical correction.

2.4. Membrane Exchange and Gas Depletion

Neutral carbon transfer through the membrane supplies DIC to the liquid without a net alkalinity flux. Define J_{abs} as positive into the liquid and express it per unit inner membrane area. With radial flux positive outward, the boundary conditions for the two totals are

$$\mathbf{J}_{y,r}(r_i, z) = \begin{pmatrix} -J_{\mathrm{abs}} \\ 0 \end{pmatrix}, \qquad \mathbf{J}_{y,r}(0, z) = \mathbf{0}.$$

At the axis, the solution is bounded and radially symmetric. At the inlet, $\mathbf{y}(r, 0, t) = \mathbf{y}_{\text{tank}}(t)$ for a uniformly mixed feed. The outlet composition follows from axial marching.

Henry's law converts the local gas CO_2 partial pressure to an equilibrium dissolved concentration. On the inner membrane area basis, the liquid-concentration transfer law is

$$J_{abs} = K_{l,\text{in}}(Hp_{CO2} - c_{0,w})$$

Here the coefficient covers gas and membrane resistance only; radial liquid transport is resolved by the fiber model. The signed law also permits desorption.

The reduced model prescribes only DIC and alkalinity membrane fluxes. Appendix A.4 explains the effective boundary condition and its interfacial interpretation.

The equivalent gas-concentration coefficient equals the liquid-basis coefficient times the dimensionless Henry partition factor $m = HRT$, provided both use the same inner-area basis. Appendix A.4 gives the inner- and outer-area flux relation.

For cross-flow gas transport, divide the bundle into gas-flow rows and the fiber axis into strips. Let $F_{\text{CO}_2,k,j}$ and $F_{g,k,j}$ be the carbon dioxide and total gas molar rates entering row k in axial strip j. If the row contains n_k identical fibers, transfer through that strip is

$$\dot{n}_{k,j} = n_k \, 2\pi r_i \int_{z_j}^{z_{j+1}} J_{\text{abs},k}(z) \, dz.$$

The coupled gas balances are

$$F_{\text{CO}_2,k+1,j} = F_{\text{CO}_2,k,j} - \dot{n}_{k,j}, \qquad F_{g,k+1,j} = F_{g,k,j} - \dot{n}_{k,j}.$$

Only CO_2 is removed, so inert-gas molar flow is conserved. The local gas concentration follows from the strip composition: $C_g = (F_{\text{CO}_2}/F_g)P/(RT)$. Gas depletion and membrane transfer are solved together.

The same transferred amount enters the liquid wall condition and depletes gas within each row-strip. The coupled calculation evaluates local gas composition from the row-strip outlet.

2.5. Recirculating Tank Balance

Define the flow-weighted cross-sectional state as

$$\bar{\mathbf{y}}(z) = \frac{2\pi}{q_f} \int_0^{r_i} v_z(r)\mathbf{y}(r, z) r \, dr.$$

Integrating the steady PDE across the lumen and along the fiber gives

$$q_f(\overline{\mathrm{DIC}}_{out} - \overline{\mathrm{DIC}}_{in}) = 2\pi r_i \int_0^L J_{\mathrm{abs}}(z)\, dz,$$

$$q_f\left(\overline{\mathrm{Alk}}_{out} - \overline{\mathrm{Alk}}_{in}\right) = 0.$$

The first identity equates membrane CO_2 input to the inlet-to-outlet increase in flow-weighted DIC. The second shows that zero alkalinity transfer leaves flow-weighted outlet Alk unchanged.

For a constant-volume tank, use the flow-weighted outlet from all returning fibers:

$$V_{\mathrm{L}} \frac{d\mathbf{y}_{\mathrm{tank}}}{dt} = Q_{\mathrm{l}}(\bar{\mathbf{y}}_{\mathrm{out}} - \mathbf{y}_{\mathrm{tank}}), \quad \mathbf{y}_{\mathrm{tank}}(0) = \mathbf{y}_{\mathrm{tank},0}.$$

The module outlet is the liquid-flow-weighted mixture from all rows. DIC and alkalinity are mixed before pH is reconstructed. Zero alkalinity wall flux keeps tank alkalinity constant under the quasi-steady coupling, even though local alkalinity can vary within a fiber.

Module uptake is the membrane flux integrated over all fibers. Additional carbon inventory is tank volume times the increase in DIC from that case's initial state; with fiber and line storage neglected, it equals time-integrated net uptake.

3. Computational Setup and Numerical Verification

3.1. Reference Case and Comparison Case

The two calculations use the same 20 × 20 array and differ only in initial DIC. The nominal inner diameter and wall thickness come from the Q3/2 data sheet [4]. Active length, tank volume, flows, initial composition, and the external coefficient are prescribed case inputs. Appendix C gives their provenance.

Geometry and inventory: The module contains 20 gas-flow rows with 20 identical fibers per row, giving 400 fibers in total. The inner and outer radii are 0.300 and $0.500\,\mathrm{mm}$, respectively; the outer radius is derived from the nominal wall thickness [4]. The selected active length is $0.200\,\mathrm{m}$ and the mixed-tank volume is $0.500\,\mathrm{L}$.

Operating conditions: Total liquid flow is $271.434\,\mathrm{mL\,min^{-1}}$ and is divided equally among all fibers. Gas flow is $234\,\mathrm{L\,min^{-1}}$ at $298.15\,\mathrm{K}$ and $101325\,\mathrm{Pa}$, with an inlet CO_2 mole fraction of $400\,\mathrm{ppm}$. Temperature and pressure remain fixed.

Initial composition: The fresh reference liquid has zero initial DIC and an alkalinity of $100\,\mathrm{mol\,m^{-3}}$. The comparison retains the same alkalinity and uses an initial DIC of $50\,\mathrm{mol\,m^{-3}}$. The reconstructed initial pH values are 13.000 and 11.500, respectively.

Transport and numerical settings: The external coefficient is $1.00 \times 10^{-4}\,\mathrm{m\,s^{-1}}$ on a liquid-concentration and inner-area basis, representing gas plus membrane resistance. The reported lumen mesh has 12 radial cells and 32 axial intervals, and the reporting horizon is $48\,\mathrm{h}$. Section 3.3 and Appendix B.2 report the associated numerical checks.

3.2. Numerical Solution

The solver uses annular finite volumes in each representative lumen and first-order implicit marching along the fiber. Radial species fluxes satisfy zero current and are projected onto DIC and alkalinity; a wall state closes the membrane exchange. Appendix B.1 gives the discrete construction.

At each axial station, Newton iteration solves the radial and wall states together with the coupled membrane and row-strip gas balances. Gas rows are evaluated in sequence, while all fibers receive the current mixed-tank composition.

An adaptive RK23 method advances the mixed-tank balance, using a direct module solve at each step. A separate gas-loss inventory provides an independent carbon-balance check. Time-integration controls and reporting times are documented in Appendix B.2.

3.3. Verification of the Coupled Calculation

Verification covers chemical equilibrium and charge flux, manufactured spatial fields, mesh and time sensitivity, and coupled carbon conservation. Spatial mesh comparisons retain all 20 gas rows and use the saved 0, 24, and $48\,\mathrm{h}$ tank states of both cases; time sensitivity is assessed over the full $48\,\mathrm{h}$ trajectory. Methods and diagnostics are given in Appendix B.2.

Manufactured-field error decreases under joint mesh refinement, with a final observed order of 0.924. Separate radial and axial refinement changes the device absorption rate by at most 0.00690%, below the 1% criterion.

Tighter time controls change tank pH by at most 7.24×10^{-6}, below the 0.02-unit criterion; both cases pass the carbon-balance checks.

4. Results and Discussion

4.1. Time Dependent Response of the Reference Device

Figure 2 shows the 48 h response of the reference liquid, initially at zero DIC and $100\,\mathrm{mol\,m^{-3}}$ alkalinity. The panels report tank pH, additional carbon uptake and instantaneous module absorption rate from the same saved calculation.

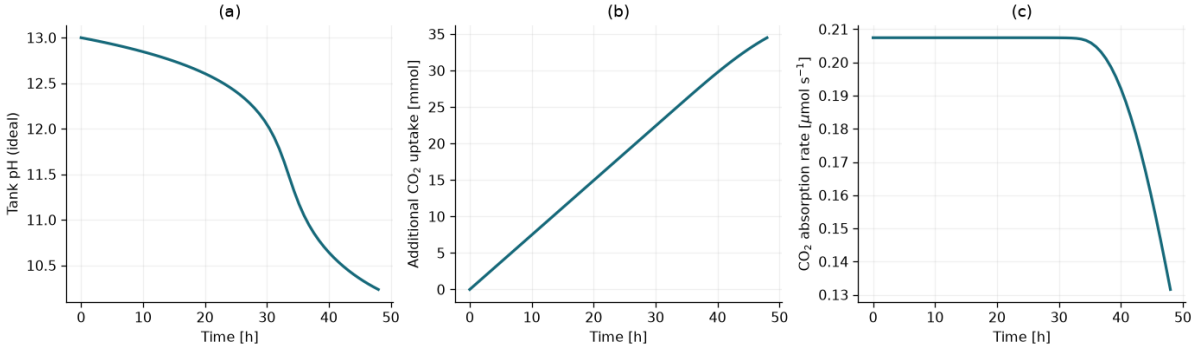


Figure 2. Time Response of the Reference 20 × 20 Array. (a) Ideal tank pH. (b) Additional cumulative CO_2 uptake relative to the initial liquid inventory. (c) Instantaneous module absorption rate. The initial DIC is zero, the initial alkalinity is $100\,\mathrm{mol\,m^{-3}}$, and the remaining inputs are summarized in Section 3.1 and Appendix C.

Tank pH decreases from 13.000 to 10.236 as additional uptake reaches 34.505 mmol at 48 h. The absorption rate remains close to $0.2074\,\mu mol\,\mathrm{s^{-1}}$ through the early part of the run and falls to $0.1316\,\mu mol\,\mathrm{s^{-1}}$ at 48 h. Accordingly, cumulative uptake is initially almost linear and bends downward as the rate decreases. The pH trajectory is more strongly curved because pH is a nonlinear equilibrium function of DIC at fixed alkalinity.

4.2. Spatial Interpretation of the Device Response

Figure 3 shows the reference state at 24 h. Gas passes through rows 1, 10 and 20 in sequence; liquid enters their fibers in parallel from the common tank. In row 10, sampled pH ranges from 12.4532 to 12.4559, a span of 0.00274. Lower values occur near the outer sampled radius and farther downstream.

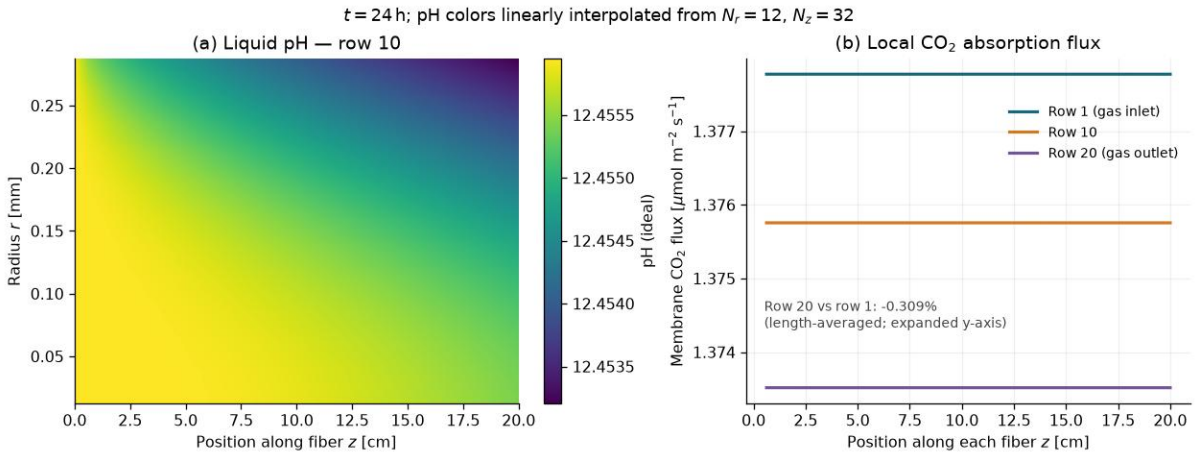


Figure 3. Reference Array at 24 h. (a) Row-10 liquid pH, linearly interpolated for display within the saved cell-center domain from 12 radial cells and 32 axial intervals. (b) Local CO_2 flux per inner membrane area along fibers in gas-flow rows 1, 10 and 20. The expanded vertical scale shows the small row effect.

Mean flux falls from $1.3778\,\mu mol\,m^{-2}\,s^{-1}$ in row 1 to $1.3735\,\mu mol\,m^{-2}\,s^{-1}$ in row 20, a 0.309% difference. The row-to-row decline is consistent with modest CO_2 depletion across the gas-flow sequence. The nearly flat profiles separately indicate little axial variation within each row at 24 h.

4.3. Effect of the Initial Absorption State

Figure 4 compares the reference liquid with one initially containing $50\,\mathrm{mol\,m^{-3}}$ DIC at the same alkalinity. Additional uptake is measured from each case's own initial carbon inventory. The initially carbonated liquid approaches a lower-pH plateau sooner and absorbs less additional CO_2 over $48\,\mathrm{h}$. Table 2 gives the initial and final values.

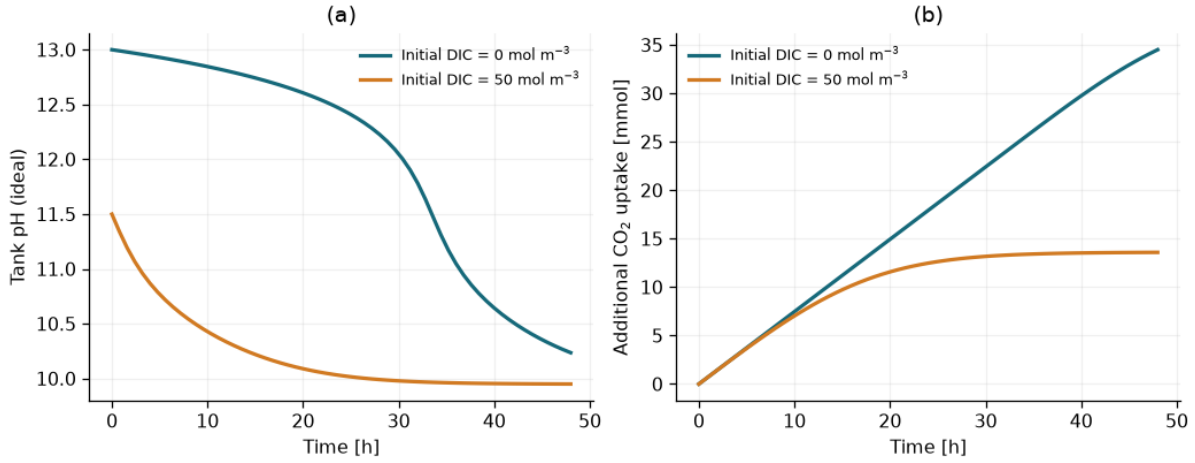


Figure 4. Effect of Initial DIC at Fixed Initial Alkalinity of $100\,\mathrm{mol\,m^{-3}}$. (a) Ideal tank pH. (b) Additional CO_2 uptake for initial DIC values of 0 and $50\,\mathrm{mol\,m^{-3}}$. Each uptake curve excludes its own initial carbon inventory. Geometry, flow rates, temperature and transport properties are identical between cases.

Table 2. Initial States and 48 h Outcomes for the Two Device Cases.

Quantity	Reference	Initial-DIC case
Initial DIC ($\mathrm{mol\,m^{-3}}$)	0	50
Initial alkalinity ($\mathrm{mol\,m^{-3}}$)	100	100
Initial pH	13.000	11.500
Tank pH at $48\,\mathrm{h}$	10.236	9.950
Additional CO_2 uptake at $48\,\mathrm{h}$ (mmol)	34.505	13.587

4.4. Applicability and Limitations

These results depend on ideal equilibrium chemistry, prescribed transport properties, an effective membrane condition, equal conditions within each gas row, and quasi-steady fiber transport. The external transfer coefficient is uncalibrated. Experimental pH and uptake data are needed to assess the predictions for an assembled contactor.

5. Conclusions

The model maps device geometry, operating conditions, and initial liquid composition to spatial absorption fields, cumulative CO_2 uptake, and tank pH. DIC and alkalinity link local equilibrium to charge-compatible transport; membrane exchange couples the liquid, gas, and tank balances.

Under the prescribed reference conditions, additional uptake is $34.505\,\mathrm{mmol}$ and tank pH falls from 13.00 to 10.236 over $48\,\mathrm{h}$. At $24\,\mathrm{h}$, the calculated spatial pH and row-to-row flux variations are small. Starting with $50\,\mathrm{mol\,m^{-3}}$ DIC reduces additional $48\,\mathrm{h}$ uptake to $13.587\,\mathrm{mmol}$. The reported conservation and refinement checks support the numerical calculation; physical validation remains to be done.

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Appendix A. Chemical and Transport Derivations

A.1. Reaction Invariant Balances

The reaction columns represent the first carbonate dissociation, the second carbonate dissociation, and water dissociation. Water is the solvent and is not a transported entry in the vector. Their stoichiometric matrix is

$$S = \begin{pmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad BS = 0.$$

For species reaction rates collected in $\boldsymbol{\rho}$, the balance and its invariant projection are

$$\partial_t \mathbf{c} + \nabla \cdot (\mathbf{v}\mathbf{c} + \mathbf{j}) = S\boldsymbol{\rho},$$

$$\partial_t \mathbf{y} + \nabla \cdot (\mathbf{v}\mathbf{y} + \mathbf{J}_y) = 0, \quad \mathbf{J}_y = B\mathbf{j}.$$

The identity $BS = 0$ removes internal reaction sources exactly for these reactions, regardless of their rates. Local equilibrium is the separate approximation described in Section 2.2. Transport can redistribute both totals spatially, and membrane carbon transfer changes DIC. Potassium has its own reaction-free balance. Bulk electroneutrality subsequently relates that additional invariant to alkalinity.

A.2. Chemical Root and Equilibrium Derivatives

For $\mathrm{DIC} \geq 0$, finite Alk, and positive equilibrium constants, the chemical equilibrium equation in Section 2.2 has exactly one positive solution. Its derivative is

$$g_h = -\mathrm{DIC}K_1 \frac{h^2 + 4K_2 h + K_1 K_2}{(h^2 + K_1 h + K_1 K_2)^2} - \frac{K_w}{h^2} - 1 < 0.$$

Moreover, $g(h) \to +\infty$ as $h \to 0^+$ and $g(h) \to -\infty$ as $h \to \infty$. Continuity and strict monotonicity establish existence and uniqueness. The potassium-only physical model further restricts alkalinity to positive values. Once the unique hydrogen concentration is found, the species concentrations are nonnegative, with all reactive concentrations positive when total carbon is positive.

Since $g_h \neq 0$, implicit differentiation gives

$$\partial_{\mathrm{DIC}} h = -\frac{\alpha_1 + 2\alpha_2}{g_h}, \quad \partial_{\mathrm{Alk}} h = \frac{1}{g_h}.$$

The chemical mapping is differentiable in the interior of the physical domain, with a one-sided carbon derivative at zero DIC. The complete Jacobian follows.

Let $\boldsymbol{\alpha} = (\alpha_0, \alpha_1, \alpha_2)^T$ and $\mathbf{n} = (h^2, K_1 h, K_1 K_2)^T$. Differentiation gives

$$\boldsymbol{\alpha}_h = \frac{(2h, K_1, 0)^T}{d} - \frac{(2h + K_1)\mathbf{n}}{d^2}.$$

Write $\mathbf{h}_y = (\partial_{\mathrm{DIC}} h, \partial_{\mathrm{Alk}} h)$, from the implicit derivatives above. The Jacobian rows are

$$G_{0:2,:} = \boldsymbol{\alpha}(1,0) + \mathrm{DIC}\boldsymbol{\alpha}_h \mathbf{h}_y,$$

$$G_{\mathrm{OH},:} = -\frac{K_w}{h^2}\mathbf{h}_y, \quad G_{H,:} = \mathbf{h}_y, \quad G_{K,:} = (0,1).$$

The implicit derivatives satisfy the invariant and electroneutrality identities in Section A.3. For the proof of root monotonicity, differentiating the carbonate alkalinity fraction produces

$$\frac{d}{dh}\left(\frac{K_1 h + 2K_1 K_2}{d}\right) = -K_1 \frac{h^2 + 4K_2 h + K_1 K_2}{d^2}.$$

The water terms supply the remaining strictly negative contribution to the root derivative, including at zero carbon.

A.3. Charge Compatible Transport

A transport law closes the invariant balances. Independent Fickian diffusion would give $\mathbf{j}_r = -D\partial_r \mathbf{c}$, with $D = \mathrm{diag}(D_i)$, and hence

$$\mathbf{J}_{y,r} = -M_F \partial_r \mathbf{y}, \quad M_F = BDG.$$

This is an exact change of variables for the assumed Fick law. With unequal ionic diffusivities, however, the independently calculated fluxes need not satisfy zero charge flux. A counterion and an electrical constraint are therefore retained in the general formulation. The ideal dilute Nernst–Planck flux is [3]

$$j_{i,r} = -D_i\left(\partial_r c_i + \zeta_i c_i \frac{F}{RT}\partial_r \varphi\right),$$

where F is Faraday's constant, R the gas constant, T absolute temperature, and φ electric potential. In the retained radial-transport approximation, electroneutrality eliminates advective charge transport. Charge conservation gives $\partial_r(r\boldsymbol{\zeta}^T \mathbf{j}_r) = 0$. Regularity at the axis then yields $\boldsymbol{\zeta}^T \mathbf{j}_r = 0$. The zero-radial-current result follows under the approximation of negligible axial nonadvective current.

Eliminating the radial electric field gives

$$\frac{F}{RT}\partial_r\varphi = -\frac{\boldsymbol{\zeta}^T DG\partial_r\mathbf{y}}{\kappa}, \quad \kappa = \sum_i \zeta_i^2 D_i c_i > 0.$$

The mobility vector b has one component for each species: its charge number multiplied by its diffusivity and concentration. Substitution of the electric field into the species flux gives the operators defined in Section 2.3. The identities below show their compatibility with the electrical and chemical constraints.

The zero-current condition applied to the Nernst–Planck flux is

$$0 = -\boldsymbol{\zeta}^T DG\partial_r\mathbf{y} - \kappa\frac{F}{RT}\partial_r\varphi.$$

Solving this scalar equation yields the radial electric field above. Since $\boldsymbol{\zeta}^T\mathbf{b} = \kappa$, substitution gives $\boldsymbol{\zeta}^T\mathcal{L} = 0$. Differentiating electroneutrality gives $\boldsymbol{\zeta}^T G = 0$, which removes the electric-field contribution when all diffusivities are equal. Multiplication by B then yields the equal-diffusivity limit below.

Differentiating the defining identity $B\mathbf{c}(\mathbf{y}) = \mathbf{y}$ and electroneutrality yields

$$BG = I_2, \quad \boldsymbol{\zeta}^T G = \mathbf{0}^T.$$

Zero charge flux also makes potassium transport consistent with the alkalinity balance: $j_{K,r} = J_{\mathrm{Alk},r}$. Thus the counterion balance is compatible with the second projected transport equation.

The flux operator also satisfies $\boldsymbol{\zeta}^T\mathcal{L} = \mathbf{0}^T$. These identities connect the reconstructed species, charge constraint, and transported totals. If all six diffusivities equal D_0, the electric-field numerator vanishes on the electroneutral equilibrium manifold and

$$M = D_0 I_2.$$

This provides a useful exact limit. The chemical equations remain nonlinear, including at the membrane, but the interior invariant diffusion operator becomes diagonal. With uniform inlet alkalinity and zero alkalinity boundary flux, constant alkalinity is a solution of this particular limit. Alkalinity remains a transported variable in the general differential-ion formulation.

A.4. Membrane Boundary Conditions and Integral Conservation

At the membrane, the outward radial normal is positive. Combining the projected flux with the membrane conditions therefore gives

$$M_{11}\partial_r\mathrm{DIC} + M_{12}\partial_r\mathrm{Alk} = J_{\mathrm{abs}}, \quad M_{21}\partial_r\mathrm{DIC} + M_{22}\partial_r\mathrm{Alk} = 0.$$

Both entries in the alkalinity wall condition have the same sign under the outward-flux convention.

Because radial liquid transport is resolved, the external coefficient includes only gas and membrane resistance. At the same area basis, $K_{l,\mathrm{in}} = K_{\mathrm{g,in}}/m$. Inner- and outer-area fluxes satisfy $r_i J_i = r_o J_o$. The signed law permits desorption if its driving force reverses.

At a species-resolved membrane, impermeant ions have zero microscopic membrane flux. The two-invariant boundary condition is an effective closure across an unresolved interface; ionic fluxes reconstructed at its liquid-side edge do not imply ion permeation through the membrane.

Integration of the flux divergence gives the continuous carbon and alkalinity balances. In the discrete system, using one shared flux with opposite signs across each interior face preserves those balances. Local field accuracy is assessed through the refinement checks in Appendix B.

Appendix A establishes the invariant, equilibrium-root, charge-transport, and integral-balance identities used by the model.

Appendix B. Discretization and Verification Details

B.1. Coupled Discretization

The differential-ion solver advances DIC and alkalinity together in each representative lumen. Radial and axial coordinates are normalized by the inner radius and active length; concentrations are scaled by the initial alkalinity. Each finite-volume balance combines the change along the liquid flow with the difference between the two radial face fluxes. There is no distributed carbon or alkalinity source in a device run.

With radius normalized by the inner lumen radius, the exact annular area and liquid-flow weights are

$$a_i = x_{i+1/2}^2 - x_{i-1/2}^2, \quad q_i = 2a_i - \left(x_{i+1/2}^4 - x_{i-1/2}^4\right), \quad \sum_i q_i = 1.$$

Interior species fluxes use concentration differences across a face and face-averaged concentrations in the electrical term. Projection of their shared face flux onto DIC and alkalinity makes adjacent cell balances conservative.

A wall unknown is coupled across the outer half-cell to the last radial cell. Its residual is the outward liquid flux plus the inward membrane flux, with zero alkalinity transfer at the wall. The same unknown wall chemistry enters the gas balance. The row-strip outlet carbon and total molar flows are obtained by subtracting the common transferred amount from their incoming values; their ratio determines the local gas composition used in the transfer law.

The implicit gas–membrane relation gives a signed flux and a physical row-strip outlet composition while preserving inert-gas flow. Newton iterations use analytic chemical and face-flux derivatives. A nonconverged or invalid state raises an error.

B.2. Array Verification Records

For each device state, compare gas carbon loss, the membrane surface integral and liquid carbon gain. The carbon residual is the largest discrepancy of the gas or liquid rate from membrane uptake. Its relative scale is the largest magnitude of these rates with a floor of 10^{-15} mol s^{-1}. The module check accepts an absolute discrepancy no larger than 10^{-14} mol s^{-1} or a relative discrepancy within 10^{-6}. Alkalinity is checked against its liquid-flow scale. The tank check compares additional carbon inventory with integrated gas loss, using an absolute floor of 10^{-11} mol or the same relative criterion.

The device mesh study retains all 20 physical gas rows and uses the baseline tank states saved at 0, 24 and 48 h. Radial cells are doubled from 12 to 24 at fixed $N_z = 32$; axial intervals are doubled from 32 to 64 at fixed $N_r = 12$. Each refined module solve receives the same tank state as its baseline comparison. Relative uptake change is the absolute rate difference divided by the refined rate magnitude,

with a floor of 10^{-15} mol s^{-1}. Outlet pH is reconstructed after liquid-flow-weighted mixing of the totals. These comparisons assess module uptake and mixed outlet pH at the three saved tank states.

Time sensitivity is tested on the baseline spatial mesh over the full 48 h. Relative tolerance decreases from 2×10^{-4} to 2×10^{-5}, while absolute concentration tolerance decreases from 10^{-6} mol m^{-3} to 10^{-7} mol m^{-3}; the maximum step decreases from 3600 to 1800 s. Differences are evaluated at the 61 common reporting times. The maximum absolute rate difference is normalized by the maximum refined rate magnitude, and the maximum accumulated-uptake difference by the maximum refined accumulated uptake, each with a 10^{-15} floor in its respective units. Table B1 summarizes these checks. The 0.02 pH-unit and 1% uptake criteria are numerical comparison criteria.

Table B1. Verification Diagnostics for the 20-Row Calculations.

Diagnostic	Reference	Initial-DIC case
Maximum radial-refinement rate change (%)	0.00259	0.00635
Maximum axial-refinement rate change (%)	0.00291	0.00690
Maximum spatial outlet-pH change	3.09×10^{-8}	1.99×10^{-8}
Maximum time-refinement tank-pH change	2.44×10^{-6}	7.24×10^{-6}
Time-refinement rate change (%)	0.000331	0.00213
Time-refinement accumulated-uptake change (%)	9.72×10^{-5}	0.000771
Tank carbon discrepancy (mol)	8.26×10^{-16}	8.36×10^{-11}
Relative tank carbon discrepancy	2.39×10^{-14}	6.15×10^{-9}
Maximum relative module carbon discrepancy	9.69×10^{-13}	2.17×10^{-7}
Configured checks	Passed	Passed

Both cases pass the local, spatial, time and carbon-conservation checks. Independent pH reconstruction differs by at most 6.22×10^{-15} in the tested states. The largest relative module carbon discrepancy is 2.17×10^{-7}, below the 10^{-6} criterion.

The manufactured-field test uses normalized coordinates $x = r/r_i$ and $s = z/L$. Its dimensionless carbon field is $\mathrm{DIC_M}(x, s) = 1 + \mathrm{e}^s(1 + x^2)$. The equal-diffusivity case uses $\mathrm{Alk_M} = 1$; the differential-ion case uses $\mathrm{Alk_M}(x, s) = 2 + 0.2\mathrm{e}^s(1 + 0.5x^2)$, thereby exercising both transported totals. Continuous flux divergence is evaluated with an independent five-point derivative of step 10^{-4} and integrated using eight-point annular quadrature. The plotted outlet error is the annular-area-weighted L_2 norm over both totals.

On meshes $(N_r, N_z) = (6, 24), (12, 48)$ and $(24, 96)$, the differential-ion error decreases from 0.04192 to 0.02303 and 0.01214. The final observed order is 0.924, consistent with first-order axial marching under joint refinement. The corresponding device-rate sensitivities for both cases are summarized in Table B1.

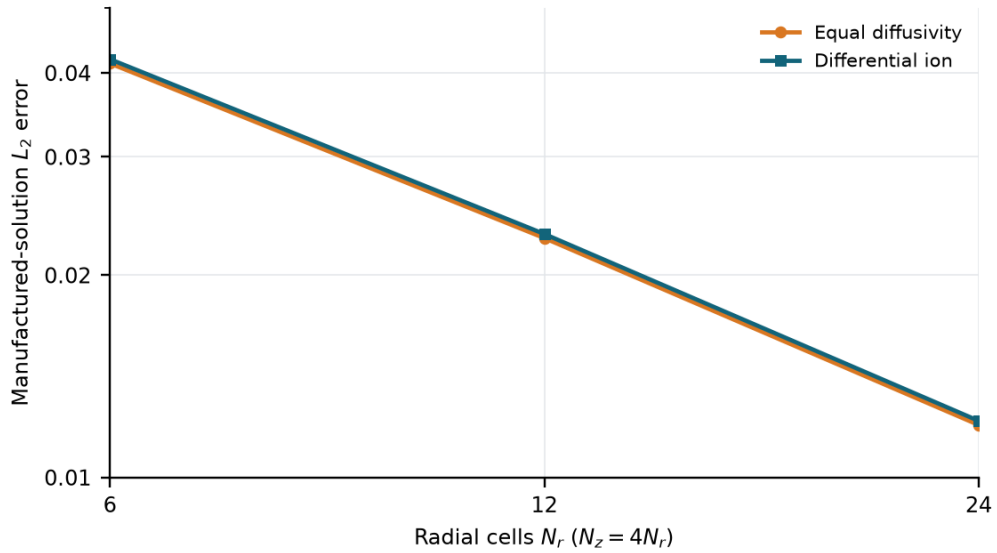


Figure B1. Manufactured-Field Convergence under Joint Spatial Refinement. The two curves show the annular-area-weighted outlet error for equal-diffusivity and differential-ion transport on meshes with four axial intervals per radial cell. Device-rate sensitivities for both initial states are reported in Table B1.

Appendix C. Parameter Values and Provenance

Properties use ideal concentration conventions at $298.15\,\mathrm{K}$; ionic-strength corrections are omitted.

At $298.15\,\mathrm{K}$, the first and second carbonate dissociation constants are $4.45 \times 10^{-4}\,\mathrm{mol\,m^{-3}}$ and $4.69 \times 10^{-8}\,\mathrm{mol\,m^{-3}}$, respectively [5,6]. The water-ionization constant is $1.00 \times 10^{-8}\,\mathrm{mol^2\,m^{-6}}$ on the stated ideal concentration basis [7]. The Henry solubility coefficient is $3.40 \times 10^{-4}\,\mathrm{mol\,m^{-3}\,Pa^{-1}}$ [8].

Concentrations are expressed in $\mathrm{mol\,m^{-3}}$. For example, $10^{-14}\,(\mathrm{mol\,L^{-1}})^2 = 10^{-8}\,(\mathrm{mol\,m^{-3}})^2$; this conversion applies to concentration-form constants.

In the order dissolved CO_2, bicarbonate, carbonate, hydroxide, hydrogen and potassium, the limiting diffusivities are $(1.92, 1.18, 0.955, 5.27, 9.31, 1.96) \times 10^{-9}\,\mathrm{m^2\,s^{-1}}$ [9]. The common factor and units apply to all six values. The external coefficient of $1.00 \times 10^{-4}\,\mathrm{m\,s^{-1}}$ is author-prescribed and uncalibrated.

The nominal $600\,\mu\mathrm{m}$ inner diameter and $200\,\mu\mathrm{m}$ wall thickness come from the Q3/2 data sheet [4]. Length, tank volume, flows and initial composition are prescribed inputs. Conversions and provenance are recorded in docs/parameters.md.

The nominal mean lumen velocity is $0.0400\,\mathrm{m\,s^{-1}}$, inner membrane area $0.150796\,\mathrm{m^2}$, and omitted lumen volume $22.62\,\mathrm{mL}$. Advective residence and slowest radial diffusion times are about 5 and $94\,\mathrm{s}$.

The companion hfmc-2d-report package retains the input configuration, numerical records and verification results; its README documents execution.